

PRODUCT SALES ANALYSIS USING MYSQL

DATASET USED- [SALES_DATA](#)

```
Create table sales(  
Product_id int,  
Sales_date date,  
Sales_rep varchar(50),  
Region varchar(50),  
Sales_amount decimal,  
Quantity_sold int,  
Product_category varchar(50),  
unit_cost double,  
unit_price double ,  
Customer_type varchar(50),  
Discount double,  
Payment_method varchar,  
Sales_channel varchar(50),  
Region_and_sales_rep  
);
```

```
Select * from sales;
```

-- A. KPI'S

-- Q1. Display all records.

```
select * from sales;
```

product_id	sale_date	Sales_Rep	Region	Sales_Amount	quantity_sold	Product_Category	unit_cost	unit_price	Customer_Type	discount	Payment_Method	Sales_Channel	R
1052	2023-02-03	Bob	North	5053.97	18	Furniture	152.75	267.22	Returning	0.09	Cash	Online	No
1093	2023-04-21	Bob	West	4384.02	17	Furniture	3816.39	4209.44	Returning	0.11	Cash	Retail	Wi
1015	2023-09-21	David	South	4631.23	30	Food	261.56	371.4	Returning	0.2	Bank Transfer	Retail	So
1072	2023-08-24	Bob	South	2167.94	39	Clothing	4330.03	4467.75	New	0.02	Credit Card	Retail	So

-- 2.Show only Sales_Rep, Region, and Sales_Amount.

```
select sales_rep,region, sales_amount from sales;
```

	sales_rep	region	sales_amount
▶	Bob	North	5053.97
	Bob	West	4384.02
	David	South	4631.23
	Bob	South	2167.94
	Charlie	East	3750.20
	Charlie	West	3761.15

-- 3.Find total number of sales.

```
select count(product_id) from sales;
```

	count(product_id)
▶	1000

-- 4.Find total sales amount.

```
select sum(sales_amount) from sales;
```

	sum(sales_amount)
▶	5019265.23

-- 5.Find average sales amount.

```
select avg(sales_amount) from sales;
```

	avg(sales_amount)
▶	5019.265230

-- 6.Find maximum and minimum sales amount.

```
select max(sales_amount),min(sales_amount) from sales;
```

	max(sales_amount)	min(sales_amount)
▶	9989.04	100.12

-- 7.Count sales by region.

```
select region,count(*) as total_sales  
from sales  
group by region;
```

	product_category	total_sales
▶	Furniture	260
	Food	226
	Clothing	268
	Electronics	246

-- 8.Count unique sales representatives.

```
select count(distinct(Sales_Rep)) as total_sales from sales;
```

	total_sales
▶	1000

-- 9.Display all sales where Sales_Amount > 5000.

```
select * from sales
where sales_amount>5000;
```

product_id	sale_date	Sales_Rep	Region	Sales_Amount	quantity_sold	Product_Category	unit_cost	unit_price	Customer_Type	discount	Payment_Method	Sales_Channel	R
1052	2023-02-03	Bob	North	5053.97	18	Furniture	152.75	267.22	Returning	0.09	Cash	Online	No
1087	2023-01-06	Eve	South	7698.92	46	Furniture	3702.51	3964.65	New	0.12	Bank Transfer	Online	So
1075	2023-10-09	Charlie	West	8239.58	18	Clothing	2228.35	2682.34	New	0.13	Bank Transfer	Online	We
1088	2023-11-16	Eve	North	8518.45	13	Furniture	2440.11	2517.6	New	0.23	Bank Transfer	Retail	No
1024	2023-11-11	Eve	West	6607.80	21	Food	622.01	641.09	Returning	0	Bank Transfer	Retail	We

-- 10.Find total sales by region.

```
select region,sum(sales_amount) from sales
group by region;
```

	region	sum(sales_amount)
▶	North	1369612.51
	West	1235608.93
	South	1154250.86
	East	1259792.93

-- 11.Find average sales by product category.

```
select product_category,avg(sales_amount) from sales
group by product_category;
```

product_category	avg(sales_amount)
Furniture	4848.144962
Food	5317.582035
Clothing	4901.023731
Electronics	5054.876585

-- 12.Find total quantity sold by each sales representative.

```
select sales_rep,sum(quantity_sold) from sales
group by sales_rep;
```

	sales_rep	sum(quantity_sold)
▶	Bob	4977
	David	6042
	Charlie	4217
	Eve	5287
	Alice	4832

-- 13. Find the top 5 highest sales.

```
select * from sales
order by sales_amount desc
limit 5;
```

	sales_amount
▶	9989.04
	9976.52
	9972.66
	9972.11
	9961.96

-- 14. Find the sales representative with the highest total sales.

```
select sales_rep, sum(sales_amount) from sales
group by Sales_Rep
order by sum(sales_amount) desc;
```

	sales_rep	sum(sales_amount)
*	David	1141737.36
	Bob	1080990.63
	Eve	970183.99
	Alice	965541.77
	Charlie	860811.48

-- 15. Find total discount given by each region.

```
select region, sum(discount) from sales
group by region;
```

	region	sum(discount)
▶	North	39.92999999038875
	West	36.18999997712672
	South	33.519999980926514
	East	42.74999991059303

-- 16.Find average unit price by product category.

```
select product_category,avg(unit_price) from sales
group by product_category;
```

	product_category	avg(unit_price)
▶	Furniture	2734.0653434753417
	Food	2657.811719371154
	Clothing	2721.2563871696816
	Electronics	2795.207191839451

-- 17.Find monthly sales.

```
select distinct(month(sale_date)), monthname(sale_date),sum(sales_amount) from sales
group by month(sale_date),monthname(sale_date)
order by month(sale_date) asc;
```

	(month(sale_date))	monthname(sale_date)	sum(sales_amount)
1	1	January	495420.37
2	2	February	368919.36
3	3	March	402638.77
4	4	April	438992.61
5	5	May	389078.76
6	6	June	418458.34

-- 18.Find yearly sales.

```
select distinct(year(sale_date)),sum(sales_amount) from sales
group by year(sale_date)
order by year(sale_date) asc;
```

	(year(sale_date))	sum(sales_amount)
▶	2023	4999937.22
	2024	19328.01

-- 19.Find total sales by customer type.

```
select customer_type, sum(sales_amount) from sales
group by customer_type;
```

	customer_type	sum(sales_amount)
▶	Returning	2513006.93
	New	2506258.30

-- 20.Find total sales by payment method.

```
select payment_method ,sum(sales_amount) from sales
group by payment_method;
```

	payment_method	sum(sales_amount)
▶	Cash	1544338.54
	Bank Transfer	1717363.17
	Credit Card	1757563.52

-- 21.Find total sales through each sales channel.

```
select sales_channel,sum(sales_amount) from sales
group by sales_channel;
```

Result Grid Filter Rows:		
	sales_channel	sum(sales_amount)
▶	Online	2458833.93
	Retail	2560431.30

-- 22.Find region-wise average discount.

```
select region, avg(discount)*100 from sales
group by region;
```

	region	avg(discount)*100
▶	North	14.955056176175564
	West	14.83196720374046
	South	14.83185839864005
	East	16.254752817715982

-- 23.Find the number of products sold in each category.

```
select product_category,sum(quantity_sold) from sales
group by product_category;
```

	product_category	sum(quantity_sold)
▶	Furniture	6729
	Food	5608
	Clothing	6922
	Electronics	6096

-- 24.Find the top-selling product category based on sales amount.

```
select product_category ,sum(sales_amount) from sales
group by product_category
order by sum(sales_amount) desc;
```

	product_category	sum(sales_amount)
	Clothing	1313474.36
	Furniture	1260517.69
	Electronics	1243499.64
	Food	1201773.54

-- 25.Find the top 3 sales representatives in each region.

```
select sales_rep,region ,sum(sales_amount) from sales
group by sales_rep,region
order by sum(sales_amount) desc
limit 3;
```

	sales_rep	region	sum(sales_amount)
▶	David	South	311416.99
	Bob	East	309876.11
	Eve	North	304172.20

-- 26.Rank sales representatives based on total sales.

```
SELECT
    sales_rep,
    total_sales,
    RANK() OVER (ORDER BY total_sales DESC) AS sales_rank
FROM (
    SELECT
        sales_rep,
        SUM(sales_amount) AS total_sales
    FROM sales
    GROUP BY sales_rep
) AS t;
```

	sales_rep	total_sales	sales_rank
▶	David	1141737.36	1
	Bob	1080990.63	2
	Eve	970183.99	3
	Alice	965541.77	4
	Charlie	860811.48	5

-- 27. Find the percentage contribution of each region to total sales.

select Region, (sum(Sales_Amount)/(SELECT SUM(Sales_Amount) FROM sales))*100 from sales
group by region;

	Region	(sum(Sales_Amount)/(SELECT SUM(Sales_Amount) FROM sales))*100
▶	North	27.287112
	West	24.617327
	South	22.996411
	East	25.099150

-- 28. Find the cumulative monthly sales.

```
SELECT  
    MONTHNAME(sale_date) AS Month,  
    SUM(sales_amount) AS Monthly_Sales,  
    SUM(SUM(sales_amount)) OVER (  
        ORDER BY MONTH(sale_date)  
    ) AS Cumulative_Sales  
FROM sales  
GROUP BY MONTH(sale_date), MONTHNAME(sale_date)  
ORDER BY MONTH(sale_date);
```

	Month	Monthly_Sales	Cumulative_Sales
▶	January	495420.37	495420.37
	February	368919.36	864339.73
	March	402638.77	1266978.50
	April	438992.61	1705971.11
	May	389078.76	2095049.87
	June	418458.34	2513508.21

-- 29. Find the month with the highest sales.

select month(sale_date), monthname(sale_date), sum(sales_amount) from sales
group by month(sale_date), monthname(sale_date)
order by sum(sales_amount) desc ;

	month(sale_date)	monthname(sale_date)	sum(sales_amount)
▶	1	January	495420.37
	11	November	467482.90
	10	October	460378.78
	8	August	443171.28
	4	April	438992.61
	6	June	418458.34

-- 30. Which sales representative generated the highest revenue in each region?


```

SELECT
  Region,
  Sales_Rep,
  Total_Sales
FROM (
  SELECT
    Region,
    Sales_Rep,
    SUM(Sales_Amount) AS Total_Sales,
    RANK() OVER (
      PARTITION BY Region
      ORDER BY SUM(Sales_Amount) DESC
    ) AS rnk
  FROM sales
  GROUP BY Region, Sales_Rep
) t
WHERE rnk = 1;

```

	Region	Sales_Rep	Total_Sales
►	East	Bob	309876.11
	North	Eve	304172.20
	South	David	311416.99
	West	Bob	286449.28

-- 31.Which product category has the highest profit?

```

select product_category,max((Unit_Price - Unit_Cost) * Quantity_Sold) as highest_revenue from sales
group by product_category
order by max((Unit_Price - Unit_Cost) * Quantity_Sold);

```

	product_category	highest_revenue
►	Furniture	21322.349739074707
	Clothing	22464.121032714844
	Food	22857.043212890625
	Electronics	23441.11669921875

-- 32.Which payment method contributes the highest revenue?

```

select payment_method,sum(sales_amount) as highest_revenue from sales
group by payment_method
order by sum(sales_amount) desc;

```

	payment_method	highest_revenue
►	Credit Card	1757563.52
	Bank Transfer	1717363.17
	Cash	1544338.54

-- 33.Which customer type receives the highest average discount?

```
select customer_type,avg(discount)*100 from sales
group by customer_type
order by avg(discount) desc;
```

	customer_type	avg(discount)*100
▶	Returning	15.306451611761604
	New	15.172619020773304

-- 34.Which sales channel generates more revenue: Online or Retail?

```
select sales_channel ,sum(Sales_Amount) from sales
group by sales_channel;
```

	sales_channel	sum(Sales_Amount)
▶	Online	2458833.93
	Retail	2560431.30

-- 35.Find the top 10 most profitable transactions.

```
select
Product_ID,
Sale_Date,
Sales_Rep,
Region,payment_method,sum((unit_price-unit_cost)*quantity_sold) from sales
group by Product_ID,
Sale_Date,
Sales_Rep,
Region,payment_method
order by sum((unit_price-unit_cost)*quantity_sold)desc
limit 10;
```

Product_ID	Sale_Date	Sales_Rep	Region	payment_method	sum((unit_price-unit_cost)*quantity_sold)
1008	2023-12-08	Alice	South	Cash	23441.11669921875
1090	2023-10-15	Charlie	South	Bank Transfer	23353.921875
1066	2023-09-25	Eve	East	Credit Card	22857.043212890625
1012	2023-06-20	Alice	South	Bank Transfer	22464.121032714844
1035	2023-05-22	Alice	South	Credit Card	22108.95263671875
1060	2023-12-16	Eve	East	Bank Transfer	21649.33203125

-- 36.Calculate profit margin (%) for each product category.

```
select product_category,(sum((unit_price-unit_cost)*quantity_sold)/(select sum(Sales_Amount) from sales))*100 from
sales
group by product_category;
```

product_category	(sum((unit_price-unit_cost)*quantity_sold)/(select sum(Sales_Amount) from sales))*100
Furniture	35.452622506702056
Food	28.313068714988667
Clothing	34.12766223582635
Electronics	31.36554777437016

-- 37.Compare sales before and after discounts.

```
SELECT
SUM(Sales_Amount) AS Sales_Before_Discount,
SUM(Sales_Amount * (1 - Discount)) AS Sales_After_Discount
FROM sales;
```

Sales_Before_Discount	Sales_After_Discount
5019265.23	4248637.739251571

-- 38. Find the top-performing sales representative every month using window functions.

```
select months,
sales_rep,
Total_Sales from
(select month(sale_date) as Months, sales_rep, sum(sales_amount) as Total_Sales,
rank() over (partition by month(sale_date) order by sum(sales_amount) desc) as rnk from sales
group by month(sale_date), sales_rep) t where rnk = 1;
```

months	sales_rep	Total_Sales
1	Eve	120821.53
2	David	89291.70
3	Eve	113311.44
4	Charlie	117609.68
5	Bob	120816.87
6	Bob	112725.70

-- 39.Month-over-Month Growth %.

```
WITH monthly_sales AS (
SELECT
YEAR(sale_date) AS year,
MONTH(sale_date) AS month,
MONTHNAME(sale_date) AS month_name,
SUM(sales_amount) AS total_sales
```

```

FROM sales
GROUP BY YEAR(sale_date), MONTH(sale_date), MONTHNAME(sale_date)
)

SELECT
    year,
    month_name,
    total_sales,
    LAG(total_sales) OVER (ORDER BY year, month) AS previous_month_sales,

    ROUND(
        (
            (total_sales - LAG(total_sales) OVER (ORDER BY year, month))
            / LAG(total_sales) OVER (ORDER BY year, month)
        ) * 100,
        2
    ) AS mom_growth_percentage

FROM monthly_sales
ORDER BY year, month;

```

	year	month_name	total_sales	previous_month_sales	mom_growth_percentage
*	2023	January	476092.36	NULL	NULL
	2023	February	368919.36	476092.36	-22.51
	2023	March	402638.77	368919.36	9.14
	2023	April	438992.61	402638.77	9.03
	2023	May	389078.76	438992.61	-11.37
	2023	June	418458.34	389078.76	7.55

-- 40. Running Profit.

```

SELECT
    sale_date,
    Daily_Profit,
    SUM(Daily_Profit) OVER (
        ORDER BY sale_date
    ) AS Running_Profit
FROM (
    SELECT
        sale_date,
        SUM((unit_price - unit_cost) * quantity_sold) AS Daily_Profit
    FROM sales
    GROUP BY sale_date
) AS t
ORDER BY sale_date;

```

	sale_date	Daily_Profit	Running_Profit
*	2023-01-01	27291.53515625	27291.53515625
	2023-01-02	978.88427734375	28270.41943359375
	2023-01-03	19448.855590820312	47719.27502441406
	2023-01-04	28880.98486328125	76600.25988769531
	2023-01-05	19980.269165039062	96580.52905273438
	2023-01-06	14037.08740234375	110617.61645507812

– Project Conclusion

This project analyzed sales data using MySQL to identify revenue trends, profit margins, regional performance, customer behavior, and sales representative effectiveness. SQL concepts including joins, aggregate functions, subqueries, common table expressions (CTEs), and window functions were used to derive business insights. The analysis supports data-driven decision-making by identifying high-performing products, profitable regions, and opportunities to improve sales performance.